Systems Engineering of LLM Workstation

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Abstract

We wish to research the process of creating and use-case testing various LLM prompts, such as prompting the user for additional details relevant to their project, assisting them in resumarising and elaborating on the information stored within a workspace, and generating intelligence from a collection of notes.

Contents

Introduction

Background and Motivation

In order to build useful software…

Project Scope and Objectives

We wish to interact with Large Language Models efficiently…

Report Structure

This report is organised into six main chapters, followed by appendices:

Chapter I: Introduction provides the background and motivation for this project, defines the scope and objectives, and outlines the structure of the report.

Chapter II: Foundations of LLM Systems establishes the theoretical groundwork by examining large language model architectures, distinguishing between inference and training workloads, characterising LLM workload requirements in terms of latency, throughput, and resource utilisation, and identifying key system integration challenges.

Chapter III: Architectural Patterns and Design explores the design space for LLM serving systems, covering inference serving architectures (single model, multi-model ensembles, and model routing), backend selection strategies (hardware, software stack, and cost-benefit considerations), query management and optimisation techniques (queuing, caching, and batch processing), and scalability patterns.

Chapter IV: Practical Implementation Considerations addresses the engineering aspects of deploying LLM systems, including infrastructure and deployment (containerisation, cloud vs. on-premise, resource provisioning), monitoring and observability (metrics, error handling, logging), and security and access control.

Chapter V: Research Findings and Analysis presents the results of our exploration, including evaluations of LLM routers and load balancers (with specific focus on the llama.cpp ecosystem and Ollama architecture), system design tradeoffs (cost vs. performance, complexity vs. maintainability, flexibility vs. stability), and lessons learned.

Chapter VI: Conclusion and Future Work summarises the key findings, provides recommendations for implementation, and identifies areas for further research.

Appendices include code examples and snippets, system diagrams and architecture charts, and research resources and references.

Foundations of LLM Systems

Understanding Large Language Models

Model Architecture Overview

Inference vs. Training Workloads

LLM Workload Characteristics

Latency Requirements

Throughput Considerations

Resource Utilization Patterns

System Integration Challenges

Architectural Patterns and Design

LLM Inference Serving Architectures

Single Model Serving

Multi-Model Ensembles

Model Routing and Load Balancing

Backend Selection Strategies

Hardware Considerations

Software Stack Evaluation

Cost-Benefit Analysis

Query Management and Optimization

Request Queuing and Prioritization

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Batch Processing Approaches

Scalability and Performance Patterns

Practical Implementation Considerations

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Containerization and Orchestration

Cloud vs. On-Premise Considerations

Resource Provisioning

Monitoring and Observability

Performance Metrics

Error Handling and Recovery

Logging and Debugging

Security and Access Control

Research Findings and Analysis

LLM Router and Load Balancer Evaluation

llama.cpp Ecosystem Exploration

Ollama Architecture Insights

Performance Comparisons

System Design Tradeoffs

Cost vs. Performance

Complexity vs. Maintainability

Flexibility vs. Stability

Lessons Learned from Exploration

Conclusion and Future Work

Summary of Key Findings

Recommendations for Implementation

Areas for Further Research

Appendices

Code Examples and Snippets

System Diagrams and Architecture Charts

Research Resources and References